

Case Study: Traffic Flow Data Visualization Using PCA and t-SNE

1. Case Background

A smart-city traffic management system continuously collects large amounts of data from different road segments. The collected data contains information about vehicle movement, traffic conditions, road occupancy, communication performance, and environmental conditions.

In the given dataset, there are 195,714 observations from 500 road segments. Each observation contains 24 numerical traffic-related features. The traffic conditions are represented using four labels:

- Free-flow
- Moderate
- Heavy
- Gridlock

The large number of features makes it difficult for traffic analysts to directly visualize and understand the relationships between different traffic observations.

2. Problem Statement

The traffic management team needs a way to understand the structure and patterns present in this high-dimensional traffic data.

Since the dataset contains 24 numerical features, directly visualizing the relationship between observations is difficult.

Therefore, the problem is:

How can high-dimensional traffic-flow data be reduced to a lower-dimensional representation while retaining important information, so that different traffic conditions and their patterns can be visualized and analyzed effectively?

3. Proposed Solution

To address this problem, the project uses two dimensionality-reduction techniques:

1. Principal Component Analysis (PCA)

PCA is used to transform the original 24-dimensional feature space into a smaller number of principal components.

The first two principal components are used to create a 2D representation of the traffic data.

2. t-distributed Stochastic Neighbor Embedding (t-SNE)

t-SNE is used to create a **nonlinear 2D representation** of the traffic observations.

Since t-SNE is computationally more expensive for very large datasets, a representative sample of 5,000 observations is used while maintaining representation from all four traffic conditions.

4. Dataset Used

Parameter	Details
Total observations	195,714
Total columns	27
Numerical features	24
Road segments	500
Traffic conditions	4
Missing values	0
Duplicate records	0

Traffic conditions

1. Free-flow
2. Moderate
3. Heavy
4. Gridlock

5. Important Features

The dataset contains features representing different aspects of traffic and the surrounding system.

Traffic-related features

- Average speed
- Vehicle density
- Average waiting time
- Occupancy percentage
- Vehicle flow
- Queue length

Vehicle movement features

- Average acceleration
- Heading
- Speed-density ratio
- Congestion pressure
- Acceleration directionality

Communication/network features

- Channel busy ratio
- Message rate
- Average communication delay
- RSSI
- Packet loss

Environmental features

- Temperature
- Visibility
- Rain intensity
- Weather factor

These features together provide a multidimensional representation of traffic conditions.

6. PCA Implementation

The 24 numerical features are first standardized so that features with different measurement scales can contribute appropriately.

PCA is then applied to the standardized dataset.

The first two principal components produced the following result:

- PC1 = 83.68% variance
- PC2 = 8.85% variance
- PC1 + PC2 = 92.53% variance

Interpretation

The first two principal components together explain 92.53% of the variance in the standardized dataset.

Therefore, the 24-dimensional traffic data can be represented in two dimensions while retaining a large proportion of its overall variation.

7. t-SNE Implementation

Because the dataset contains **195,714 observations**, running t-SNE on the complete dataset would be computationally expensive.

Therefore, a representative sample of **5,000 observations** is selected.

The sample contains:

Traffic condition	Number of observations
Free-flow	1,477
Moderate	1,380
Heavy	1,212
Gridlock	931
Total	5,000

The sampled data is standardized and then transformed into two t-SNE dimensions:

- tSNE1

- tSNE2

8. Results and Observations

PCA Result

PCA provides a compact representation of the dataset.

The first two principal components explain 92.53% of the total variance, allowing the major variation in the traffic data to be visualized in two dimensions.

t-SNE Result

The t-SNE visualization shows the relationships between observations in a nonlinear 2D space.

From the obtained visualization:

Free-flow observations are mostly located separately from the other traffic conditions, while Heavy and Gridlock observations show overlap.

This indicates that the traffic conditions do not form completely distinct groups in the t-SNE representation.

9. PCA and t-SNE Comparison

PCA	t-SNE
Linear dimensionality reduction	Nonlinear dimensionality reduction
Focuses on variance	Focuses on local relationships
Provides explained variance	Primarily used for visualization
Applied to the complete dataset	Applied to a representative 5,000-row sample
PC1 + PC2 explain 92.53% variance	Produces a 2D embedding
Provides a global representation of variation	Helps reveal local patterns and groupings

10. Final Case Study Conclusion

The project demonstrates how dimensionality-reduction techniques can be used to understand high-dimensional traffic-flow data.

PCA successfully reduced the 24 numerical traffic features into two principal components while retaining **92.53% of the total variance**.

t-SNE provided a nonlinear 2D visualization of a representative 5,000-observation sample. The visualization showed that **Free-flow observations were mostly separated, whereas Heavy and Gridlock observations showed overlap**.

Therefore, PCA and t-SNE provide **complementary perspectives** for analyzing and visualizing high-dimensional traffic data. PCA provides a compact representation of the overall variation, while t-SNE helps explore local relationships and patterns between observations.